

Dexter Fernandes

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<https://github.com/Dexter-Fernandes>

Senior Software Engineer

Python engineer with 3+ years of experience building production AI and data systems across ingestion, training, deployment, monitoring, and internal tooling. Strong background in scalable Python pipelines, Dockerised services, Kafka-based messaging, FastAPI and Django applications, and production debugging for real-time systems. Experienced in profiling, observability, and reliability improvements across cross-functional, remote-friendly engineering environments. Looking to bring that systems mindset to AI infrastructure and platform engineering.

WORK EXPERIENCE

Esbaaar • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Built multimodal data ingestion, cleansing, training, and validation pipelines using Python, Pandas, Apache Airflow, MLflow, DVC, Optuna, and CVAT, cutting model iteration time by 40% and reducing rollout time from 3 days to under 24 hours, handling upto 3.6 million records.
- Refactored legacy services into Dockerised microservices with Kafka-based messaging, structured logging, health checks, retries, and safe restarts, improving fault isolation, increasing MTBF by 35%, and reducing live video pipeline crashes by 67%.
- Developed an internal Django application and dashboard to surface inference outputs, system health, and performance metrics, improving debugging and root-cause analysis across deployed systems.
- Diagnosed and resolved production issues in Dockerised microservices, improving service continuity through stronger logging, health checks, safe restarts, and workload isolation.
- Documented deployment guardrails, failure modes, and recovery procedures, and contributed to peer code reviews and QA practices to improve maintainability and engineering handover.
- Worked with cross-functional engineering and client teams to translate performance findings into deployment and optimisation decisions for production systems.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI • Full-time

- Owned end-to-end production ML pipelines from data ingestion and labelling through training, optimisation, deployment, monitoring, and field validation for industrial safety systems.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% while accelerating iteration cycles on large real-world datasets.
- Built low-latency perception pipelines using OpenVINO, and C++17, increasing throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel VPU hardware under tight power and memory constraints.
- Led a 4-member engineering team, set experimentation standards, reviewed performance results, and collaborated with hardware, software, and QA teams to improve production delivery.
- Evaluated edge versus cloud inference trade-offs to minimise bandwidth, latency, and operational cost while maintaining safety-critical performance, supporting production cloud-hosted ML workloads on Microsoft Azure.
- Conducted systematic quantisation and optimisation experiments benchmarking accuracy–latency–memory trade-offs and reducing model size by 50% while meeting real-time constraints.

PROJECTS

Modern Edge CV Benchmarking

Mar 2026

- Benchmarked modern computer vision models including YOLOv8, YOLOv10, and RT-DETR on a shared validation subset, comparing mAP, inference time, and model size to understand trade-offs for low-latency deployment.
- Exported models to ONNX and evaluated edge deployment paths using OpenVINO, strengthening practical understanding of optimisation and model selection for production-style CV systems.
- Extended the work by studying recent edge and foundation-model approaches including Grounding DINO, MobileSAM, Depth Anything v2, and EfficientViT to stay current with state-of-the-art computer vision methods.

End-to-End Damage Severity & Claim Assessment

Nov 2025

- Developed an end-to-end automotive claims assessment pipeline combining computer vision-based damage detection with a trained severity-classification model, achieving accurate triage for real-world vehicle damage scenarios.
- Built a lightweight repair-insight engine that transforms model outputs into insurer-friendly explanations, using structured reasoning + LLM-generated narratives to deliver clear, interpretable claim summaries.
- Designed the entire workflow (data preprocessing, model training, evaluation, and a user-facing inference module) demonstrating practical understanding of insurance claims processes.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Radiological SLAM with LiDAR Odometry. Designed and implemented a real-time SLAM system using RTAB-Map on the Clearpath Husky platform, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Evaluated RTAB-Map, LIO-SAM, and Cartographer across comparable scenarios to assess mapping and localisation behaviour in simulation before real-world deployment.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, including the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Developed and maintained ROS2 nodes using tf2 and message synchronisation, and debugged tf trees and frame connectivity.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, perception systems, and robotics-adjacent engineering.

CERTIFICATIONS

Building RAG Agents with LLMs

Mar 2026 - Apr 2026

Nvidia

Deep Learning Specialisation

Jul 2019

deeplearning.ai

SKILLS

- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, API design, AWS, CI/CD, Docker, DVC (Data Version Control), Intel VPUs, Microsoft Azure, MLFlow

- **Programming & Tooling:** Bash, C++17, CMake, colcon, Django, FastAPI, Git, Intel VTune, Linux, NumPy, Pandas, PG vector, PostgreSQL, Python, Qt, SciPy
- **Deep Learning and Optimisation:** Bayesian Optimisation, Convolutional Neural Networks, Data Augmentation, Layer fusion, MultiRocket, Object Detection & Tracking, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Scikit-Learn, Structured pruning, TensorFlow, TensorRT, YOLO
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership
- **Point Cloud & Vision Processing:** Denoising, Open3D, OpenCV, Point Cloud Library, Scikit-Image
- **Computer Vision & Image Processing:** Data augmentation, Image processing, LiDAR extrinsics calibration, Object detection, Open3D, OpenCV, PCL (Point Cloud Library), Sensor calibration
- **AI-Assisted Development:** ChatGPT Codex, Claude Code, Pompt engineering